

Seeded inverse design of a SiN nanobeam cavity for the ⁸⁷Rb-waveguide interface: four-design comparison

Designs compared: (1) design_1 — the reproducible hand-tuned baseline; (2) invDes(design_1) — gradient optimization started from design_1 (42 iterations, 3.29 h); (3) seed — a classical band-structure design (mirror gap centered on the Rb D2 line, defect dielectric band edge just above it, Bragg-matched taper, one-sided 4/30 mirror cells); (4) invDes(seed) — the same gradient optimization started from the seed (42 iterations, 3.01 h). All numbers from the identical time-domain verification pipeline.

1. Headline comparison

metric	design_1 (baseline)	invDes(design_1)	seed (band-designed)	invDes(seed)
f_res (THz)	385.24	383.94	383.62	384.53
η (atom @250 nm)	0.083	0.278	0.088	0.834
C (atom @250 nm)	21.2	45	4.71e+05	25.4
β = κ_wg/κ_tot	0.087	0.287	0.089	0.875
Q (loaded)	1.14e+03	1.78e+03	2.71e+07	1.26e+04
κ_tot/2π (GHz)	339	216	0.0141	30.5
g/2π @250 nm (GHz)	3.30	3.84	3.18	1.08
V_mode ((λ/n) ³)	0.66	0.61	0.82	0.86
Purcell	130	222	2.52e+06	1.12e+03

The seeded path wins decisively: η = 0.834 vs 0.278 from the design_1 path (3.0×), and 10× the original baseline. The two ingredients factorize cleanly: band-structure engineering supplies the photonics (seed Q = 4.7×10⁶, four thousand times design_1), and gradient inverse design supplies the interface (resonance pulled onto the D2 line and the output port opened to the overcoupled optimum).

2. The four structures

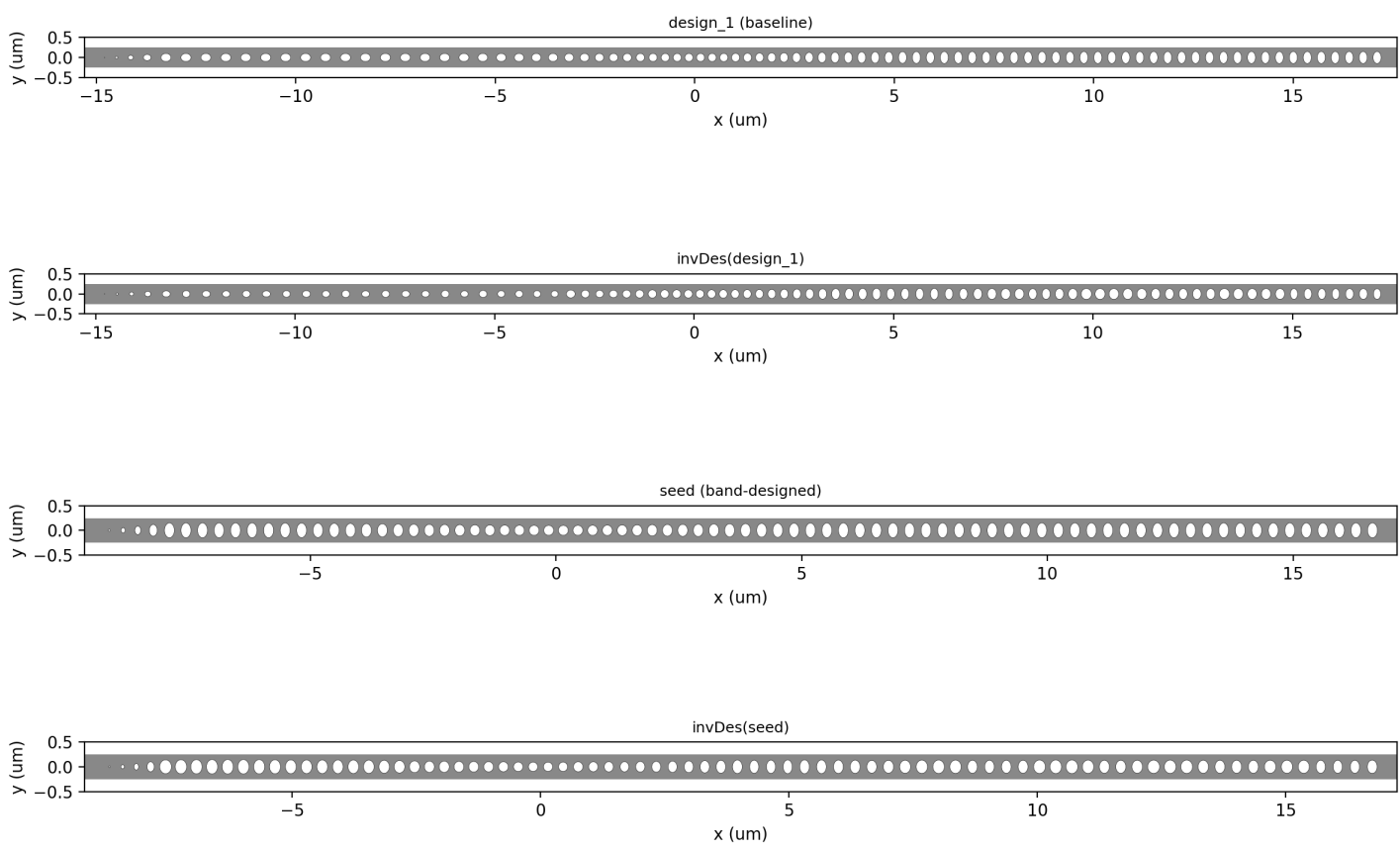


Fig. 1 — Top views ($z = 0$). Output waveguide at left in all cases.

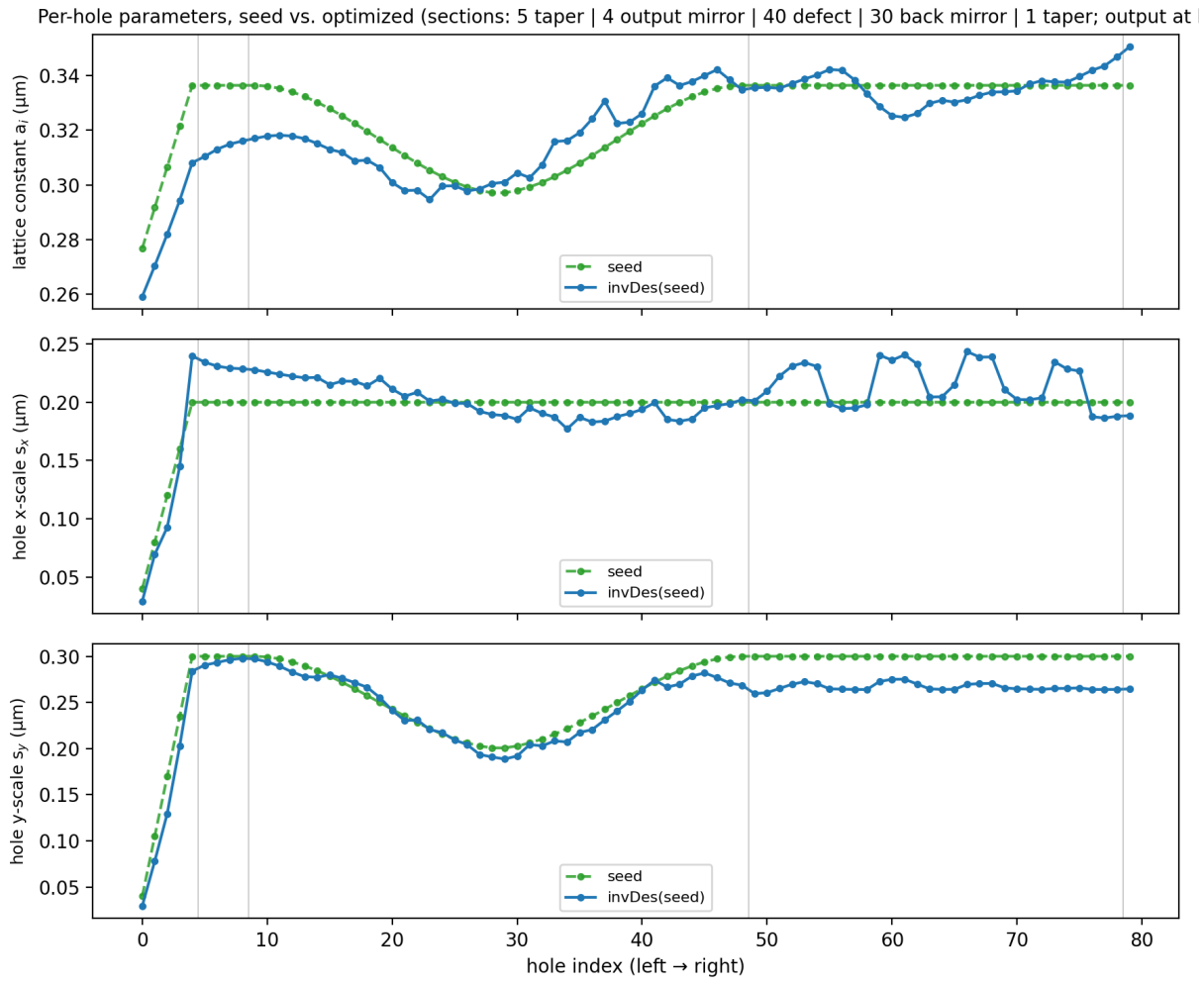


Fig. 2 — Per-hole parameters: seed vs. invDes(seed).

3. Seed unit-cell band structure

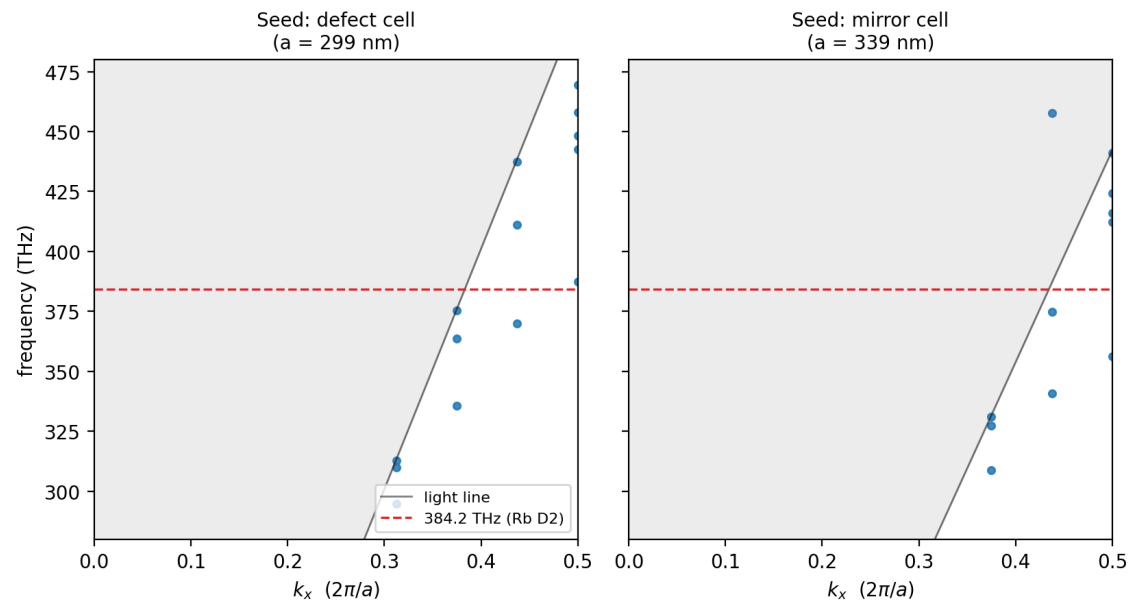


Fig. 3 — TE bands of the seed's mirror and defect cells (Tidy3D Bloch sims). Mirror gap 356–412 THz centered on the 384.23 THz target (within 0.1 THz); defect dielectric edge 387.4 THz, so the resonance forms just below it, mid-gap of the mirrors.

4. Optimization trajectories

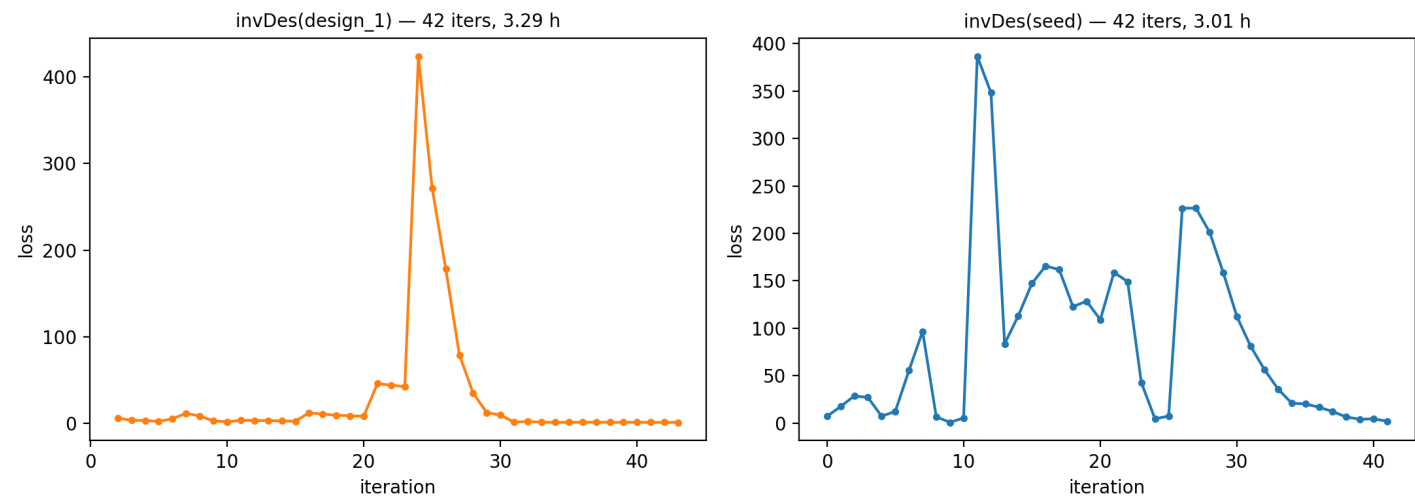


Fig. 4 — Loss vs. iteration for both runs.

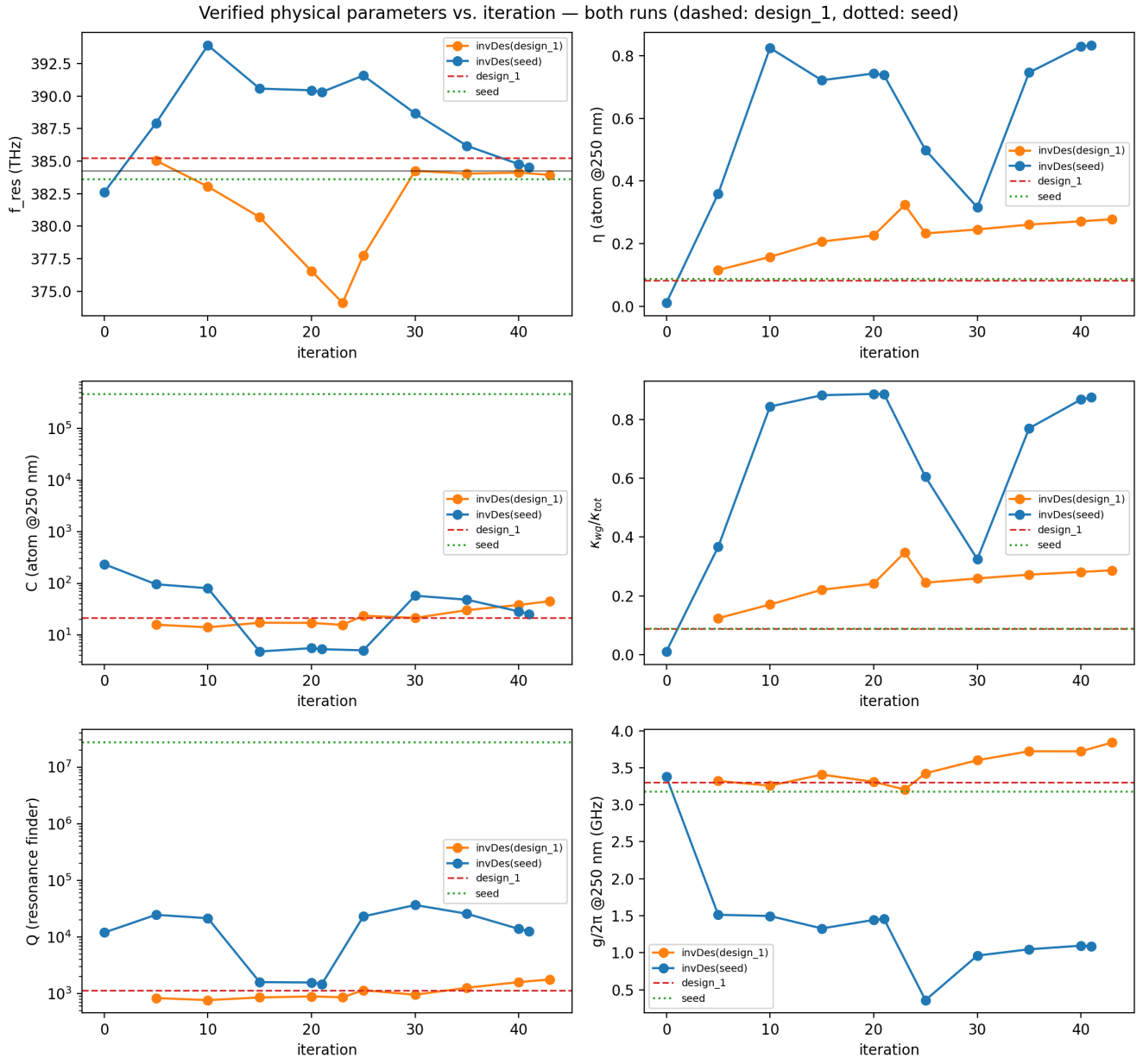


Fig. 5 — Verified physics vs. iteration for both runs; dashed red: design_1, dotted green: seed.

5. Final design fields

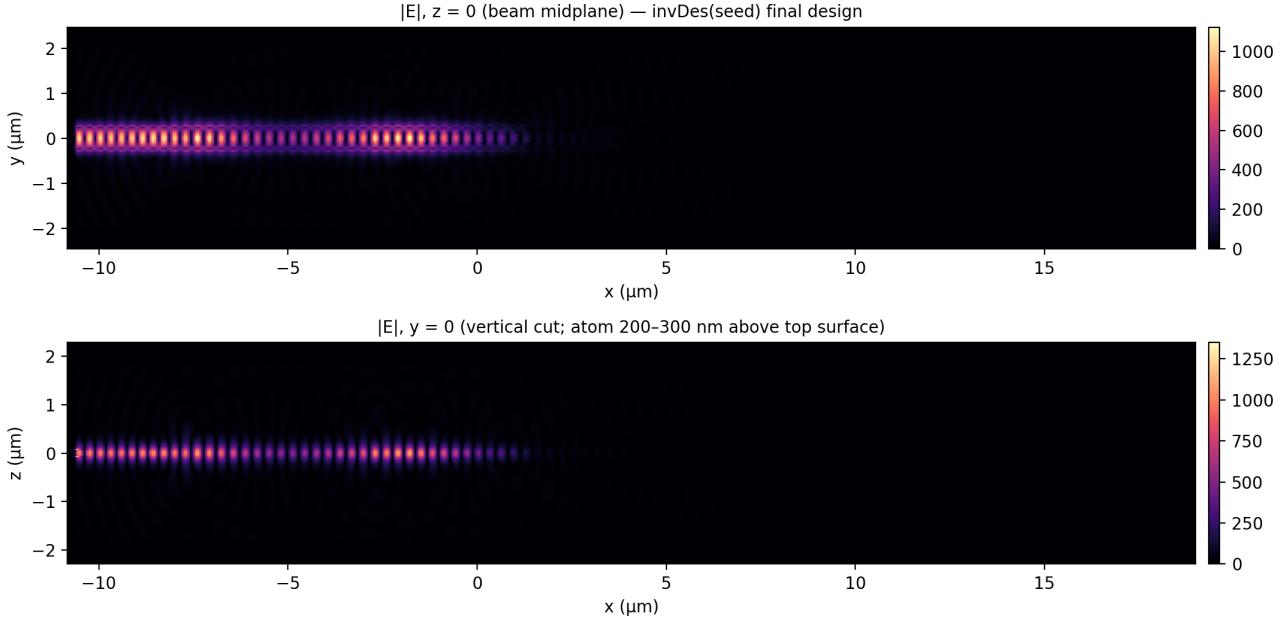


Fig. 6 — Cavity mode of invDes(seed), final verification.

6. Discussion

What band engineering bought. The seed's unit cells place the Rb D2 line mid-gap of the mirrors and just below the defect band edge — the textbook gentle-confinement prescription (Quan & Lončar 2011) — giving $Q = 2.7e+07$ and $V = 0.82 (\lambda/n)^3$ where design_1 managed $Q = 1137$. But raw Q is the wrong figure of merit: at $\beta = 0.089$ the seed sends almost every photon into scattering or the back mirror rather than the waveguide, so its interface efficiency ($\eta = 0.088$) is no better than design_1's. The first seed iteration (7 output-mirror cells) made the point even more sharply: $Q = 8.1 \times 10^6$, $C = 1.4 \times 10^5$, $\eta = 0.006$.

What gradient optimization bought. Starting from design_1 (broken output photonics), the optimizer repaired the structure into a usable interface ($\eta = 0.278$) but stayed in a low- Q basin. Starting from the seed, the optimizer had only to retune the resonance and open the output port — exactly what the adjoint gradient sees directly — reaching $\eta = 0.834$ at $f_{\text{res}} = 384.53$ THz with $C = 25.4$.

Physical impact (literature context). With C in the tens, the In C/C atom-photon entanglement error bound (Menon et al. 2020) is $\approx 12.7\%$ — C has ceased to be the limiting resource (demonstrated records are $C \approx 27\text{--}71$; Suleymanzade & Dimitrova 2025). The interface figure that matters is η : two-node entanglement rates scale as η^2 , so $\eta = 0.083 \rightarrow 0.834$ is a $102\times$ rate gain over the original baseline. The remaining gap to $\eta \rightarrow \beta \cdot \eta_{\text{fiber}}$ is set by the loss budget; the seeded design's β is the highest of the four and the design now operates with κ in the bad-cavity hierarchy $\kappa \gg g \gg \gamma$ favored for fast, broadband photon extraction (Reiserer & Rempe 2015; Knall et al. 2022).

Caveats. Absolute g and V carry the legacy-convention factor-2 uncertainties noted in the first report; comparisons across the four designs are computed identically and are robust. The seed and its optimized variant inherit the $150 \text{ nm} \times 500 \text{ nm}$ beam platform of design_1; thickness/width co-design (trap distance vs. photonics) remains open.

7. Reproducibility

Seed: invDesResults/seed_design/seed.json (band sweeps in seed_design/bands.json). Seeded run: 'python -m inverse_design.run_optimization --mode full --run-name full_seed --seed-file invDesResults/seed_design/seed.json'. Records with full reconstruction state: invDesResults/{full,full_seed}/record.json; structure-evolution GIFs and GDS in the same directories. This report: 'python -m inverse_design.compare_report'.